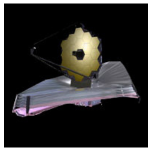

8004 - Cliff Hangers: Testing for Atmosphere-Mantle Interactions in Radius Cliff Planets

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
Observation Folder				
	1	TOI-431 d	NIRSpec Bright Object Time Series	(1) TOI-431d
	2	TOI-431 d	NIRSpec Bright Object Time Series	(1) TOI-431d
	3	HD 12572 b	NIRSpec Bright Object Time Series	(2) HD-12572b
	4	HD 12572 b	NIRSpec Bright Object Time Series	(2) HD-12572b

ABSTRACT

Remarkable progress has been made in recent years to understand the demographics of exoplanets from transit and Doppler survey data. The sharp drop in exoplanet occurrence between 2.5 Rearth and 4 Rearth, known as the Radius Cliff, is arguably the most dramatic feature in the occurrence distribution of small planets, yet, planets along the cliff remain unexplored by JWST. These "radius cliff" planets are hypothesized to have distinct elemental ratios, when compared to their larger radius neighbors, because of large-scale magma-atmosphere interactions that could modify the Carbon and Oxygen abundances, ultimately lowering the atmospheric C/O ratio. Our program will test this hypothesis by obtaining NIRSpec G395H transmission specroscopy of two radius cliff planets and answer the question: do the atmospheres of radius cliff planets trace the chemical changes predicted by theory?

Focusing on the radius cliff regime will have a far reaching out-of-field impact on theoretical models that are tuned to reproduce the position and steepness of the radius cliff. Our program will leverage JWST's unique capabilities to detect the major carbon and oxygen-bearing species in transmission for this high-priority temperature-radius regime. Combined with JWST observations of sub-Neptunes at the top of the cliff and Neptune-size planets at the bottom of the cliff, our sample will provide meaningful constraints on theoretical models of the physical processes that sculpt the frequency distributions of small planets.

OBSERVING DESCRIPTION

We will observe 2 planets around 2 stars. We will observe each planet with the same observational mode and set-up for consistency. We will observe two visits of each planet transit event. Therefore, in total we will observe 4 individual transit events to achieve the needed precision for the outlined science goals.

We use NIRSpec in BOTS (Bright Object Time Series) mode, which requires the S1600A1 aperture with a fixed 1.6”x1.6” field of view (FoV). We will use the G395H/F290LP combination (2.87-5.27 microns) and SUB2048 subarray for all of our observations. This will ensure a uniform treatment of the data. This program is schedulable throughout cycle 4, and there are multiple opportunities throughout the year where our target stars are visible to JWST, each with multiple transit opportunities. Each observation must be scheduled to cover the transit of the planet. We have set the phase constraints to allow for a range of 60 minutes for the JWST observation start window for ease of scheduling. For each of our observations we will obtain a full observation of 2*(0.75 Hr + Max(1 Hr + T14/2, T14) (the "dwell equation" recommended by ExoCTK).

We anticipate needing to discard a maximum of 0.75 hours of data at the start of each observation due to detector settling effects which result in a ramp-like increase in the flux measured which would be unusable in our analysis. Our targets have transit times of 4.913 hrs (HD 12572 b) and 3.3 hrs (TOI-431 d). We calculated the time needed for each of our observations based on our science goal of matching the precision of GJ 3470 b retrieved C/O precision for a range of atmospheric metallicity and cloud obscuration levels. In total, the time needed includes data prior to, during, and following the transit event to obtain an adequate baseline of the stellar flux.

We will measure 4 transits over 2 targets with a science time of 31.8, and a total charged time of 49 hours including the observatory resets, overheads, slewing, and target acquisition. Our two targets are bright, therefore we cannot use them for target acquisition. For TOI-431 d, we will conduct target acquisition (TA) on an alternative target to avoid saturation. For the TA we will utilize the Wide Aperture Target Acquisition (WATA) mode on a target that is within the visit splitting distance of our target stas. We selected a source with valid 2MASS J magnitudes and Gaia DR2 proper motions to ensure they are suitable for accurate TA. We used VizieR and Aladin to validate the source, and the online JWST ETC to determine the set-up required. For TOI-431 d's TA star we will use the SUB32 subarray with the CLEAR filter. For HD 12572 b so we will rely on facility pointing. We will double check both of these, particularly the star for which we have no TA star, with scheduling prior to conducting the observations to ensure success of the program.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	TOI-431d	RA: 05 33 4.6005 (83.2691688d) Dec: -26 43 28.27 (-26.72452d) Equinox: J2000	Proper Motion RA: 16.886 mas/yr Proper Motion Dec: 150.779 mas/yr Parallax: 0.0306517" Epoch of Position: 2000	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Exoplanets]				
(2)	HD-12572b	RA: 02 03 37.0253 (30.9042721d) Dec: +21 16 51.11 (21.28086d) Equinox: J2000	Proper Motion RA: 158.857 mas/yr Proper Motion Dec: 107.476 mas/yr Parallax: 0.0148601" Epoch of Position: 2000	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Exoplanets]				
(3)	TA-TOI-431d	RA: 05 33 7.1097 (83.2796237d) Dec: -26 43 49.59 (-26.73044d) Equinox: J2000	Proper Motion RA: 5.4872712544669575 mas/yr Proper Motion Dec: 7.241940424171782 mas/yr Parallax: 0.0010054152824797835" Epoch of Position: 2000	
Comments: Category=Star Description=[M stars] Extended=NO				

Observation

Proposal 8004, Observation 1: TOI-431 d

Tue Mar 11 22:01:02 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRSpec Bright Object Time Series

Diagnostics

(TOI-431 d (Obs 1)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure.

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	TOI-431d	RA: 05 33 4.6005 (83.2691688d) Dec: -26 43 28.27 (-26.72452d) Equinox: J2000	Proper Motion RA: 16.886 mas/yr Proper Motion Dec: 150.779 mas/yr Parallax: 0.0306517" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Star
Description=[Exoplanets]

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	3 TA-TOI-431d	WATA	SUB32	CLEAR	NRSRAPID	3	1	1	0.08	226404

Template

Subarray
SUB2048

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	G395H/F290LP	NRSRAPID	3	8124	1	1	8124	29477.772	

Special Requirements	
	Phase 0.9830462282545397 to 0.9863899860872924 with period 12.46103 Days and zero-phase 2458764.61739 HJD Time Series Observation No Parallel Attachments

Observation

Proposal 8004, Observation 2: TOI-431 d

Tue Mar 11 22:01:02 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRSpec Bright Object Time Series

Diagnostics

(TOI-431 d (Obs 2)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure.

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	TOI-431d	RA: 05 33 4.6005 (83.2691688d) Dec: -26 43 28.27 (-26.72452d) Equinox: J2000	Proper Motion RA: 16.886 mas/yr Proper Motion Dec: 150.779 mas/yr Parallax: 0.0306517" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Star
Description=[Exoplanets]

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	3 TA-TOI-431d	WATA	SUB32	CLEAR	NRSRAPID	3	1	1	0.08	226404

Template

Subarray
SUB2048

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	G395H/F290LP	NRSRAPID	3	8124	1	1	8124	29477.772	

Special Requirements	
	Phase 0.9830462282545397 to 0.9863899860872924 with period 12.46103 Days and zero-phase 2458764.61739 HJD Time Series Observation No Parallel Attachments

Observation

Proposal 8004, Observation 3: HD 12572 b

Tue Mar 11 22:01:02 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRSpec Bright Object Time Series

Diagnostics

(HD 12572 b (Obs 3)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure.

(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(Visit 3:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	HD-12572b	RA: 02 03 37.0253 (30.9042721d) Dec: +21 16 51.11 (21.28086d) Equinox: J2000	Proper Motion RA: 158.857 mas/yr Proper Motion Dec: 107.476 mas/yr Parallax: 0.0148601" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Star
Description=[Exoplanets]

Template

TA Method	Subarray
NONE	SUB2048

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	G395H/F290LP	NRSRAPID	6	6494	1	1	6494	41136.113	

Special Requirements

Phase 0.986571264407061 to 0.9885770869408051 with period 20.772858 Days and zero-phase 2458767.42089 HJD
Time Series Observation
No Parallel Attachments

Observation

Proposal 8004, Observation 4: HD 12572 b

Tue Mar 11 22:01:02 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRSpec Bright Object Time Series

Diagnostics

(HD 12572 b (Obs 4)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure.

(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(Visit 4:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	HD-12572b	RA: 02 03 37.0253 (30.9042721d) Dec: +21 16 51.11 (21.28086d) Equinox: J2000	Proper Motion RA: 158.857 mas/yr Proper Motion Dec: 107.476 mas/yr Parallax: 0.0148601" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Star
Description=[Exoplanets]

Template

TA Method	Subarray
NONE	SUB2048

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	G395H/F290LP	NRSRAPID	6	6494	1	1	6494	41136.113	

Special Requirements

Phase 0.986571264407061 to 0.9885770869408051 with period 20.772858 Days and zero-phase 2458767.42089 HJD
Time Series Observation
No Parallel Attachments